In [ ]:

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                                             Assignment no:5
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Div: B
Batch:B2
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In [1]:

```python
# HPC Application using TensorFlow (Parallel Training)

import tensorflow as tf
from tensorflow import keras
import numpy as np
import time

# Step 1: Start time (to measure performance)
start_time = time.time()

# Step 2: Create distributed strategy (HPC concept)
strategy = tf.distribute.MirroredStrategy()

print("Number of devices used:", strategy.num_replicas_in_sync)

# Step 3: Load dataset (MNIST)
(x_train, y_train), (x_test, y_test) = keras.datasets.mnist.load_data()

# Step 4: Normalize data
x_train = x_train / 255.0
x_test = x_test / 255.0

# Step 5: Build model inside strategy scope (important for HPC)
with strategy.scope():
    model = keras.Sequential([
        keras.layers.Flatten(input_shape=(28, 28)),
        keras.layers.Dense(128, activation='relu'),
        keras.layers.Dropout(0.2),
        keras.layers.Dense(10, activation='softmax')
    ])

    # Compile model
    model.compile(
        optimizer='adam',
        loss='sparse_categorical_crossentropy',
        metrics=['accuracy']
    )

# Step 6: Train model (parallel execution)
history = model.fit(
    x_train, y_train,
    epochs=5,
    batch_size=64,
    validation_split=0.1
)

# Step 7: Evaluate model
loss, accuracy = model.evaluate(x_test, y_test)

# Step 8: End time
end_time = time.time()
```

```python
# Step 9: Print results
print("\nTest Accuracy:", accuracy)
print("Total Training Time:", round(end_time - start_time, 2), "seconds")
```

```
INFO:tensorflow:Using MirroredStrategy with devices ('/job:localhost/replica:0/task:0/device:CPU:0',)
Number of devices used: 1
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz
11490434/11490434 ━━━━━━━━━━━━━━━━━━━━ 5s 0us/step
C:\Users\Monuu\anaconda3\Lib\site-packages\keras\src\layers\reshaping\flatten.py:37: UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.
  super().__init__(**kwargs)
WARNING:tensorflow:From C:\Users\Monuu\anaconda3\Lib\site-packages\keras\src\backend\tensorflow\core.py:232: The name tf.placeholder is deprecated. Please use tf.compat.v1.placeholder instead.

Epoch 1/5
844/844 ━━━━━━━━━━━━━━━━━━━━ 7s 7ms/step - accuracy: 0.8951 - loss: 0.3638 - val_accuracy: 0.9573 - val_loss: 0.1515
Epoch 2/5
844/844 ━━━━━━━━━━━━━━━━━━━━ 5s 6ms/step - accuracy: 0.9487 - loss: 0.1756 - val_accuracy: 0.9717 - val_loss: 0.1033
Epoch 3/5
844/844 ━━━━━━━━━━━━━━━━━━━━ 5s 6ms/step - accuracy: 0.9607 - loss: 0.1333 - val_accuracy: 0.9765 - val_loss: 0.0886
Epoch 4/5
844/844 ━━━━━━━━━━━━━━━━━━━━ 5s 6ms/step - accuracy: 0.9684 - loss: 0.1054 - val_accuracy: 0.9763 - val_loss: 0.0830
Epoch 5/5
844/844 ━━━━━━━━━━━━━━━━━━━━ 5s 6ms/step - accuracy: 0.9721 - loss: 0.0898 - val_accuracy: 0.9773 - val_loss: 0.0735
313/313 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - accuracy: 0.9761 - loss: 0.0760

Test Accuracy: 0.9761000275611877
Total Training Time: 43.47 seconds
```

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